

A PROPOSED STABILITY CRITERION FOR THE RIEMANN HYPOTHESIS BASED ON A SHARED FUNCTIONAL SYMMETRY

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ABSTRACT. This paper defines a set of constants, T and J , derived axiomatically from principles of algebraic simplicity ($a + b = 1/2$, $a/b = \phi$). These constants are shown to be unique under a convergence constraint. From these constants, we derive a potential function, $V(s)$, which is proven to be a necessary consequence of the axiomatic structure. We prove that this potential function simplifies to the identity $V(s) = s - 1/2$. Next, we prove that $V(s)$ shares an identical reflectional anti-symmetry with the derivative of the completed Zeta function, $\xi'(s)$. That is, both functions satisfy $f(s) = -f(1 - s)$. Based on this shared structural property, we propose a single, falsifiable conjecture, the 'Principle of Harmonic Stability,' which links functions with this specific anti-symmetry to the ground state of the potential. Conditional on this principle, it is proven that all non-trivial zeros of $\zeta(s)$ must lie on the critical line $\text{Re}(s) = 1/2$. The work, therefore, reframes the Riemann Hypothesis as being equivalent to the validity of the proposed Principle of Harmonic Stability, inviting further investigation into this structural criterion.

1. INTRODUCTION

The Riemann Hypothesis posits that all non-trivial zeros of the Riemann Zeta function, $\zeta(s)$, have a real part equal to $1/2$. It is often studied via the completed Zeta function, $\xi(s)$, whose zeros coincide with the non-trivial zeros of $\zeta(s)$. The function $\xi(s)$ is entire and defined as:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

where $\Gamma(s)$ is the Gamma function. The key property of $\xi(s)$ for our analysis is its functional equation, $\xi(s) = \xi(1-s)$.

The objective of this paper is not to provide an unconditional proof of the hypothesis. Rather, our goal is to derive a new, falsifiable criterion for the location of the zeros. We achieve this by first constructing a simple algebraic framework from first principles. We derive a potential function, $V(s)$, that arises as a necessary consequence of this framework's structure. We then demonstrate that this potential shares a key reflectional symmetry with the derivative of $\xi(s)$. This shared symmetry motivates the proposal of a new stability principle. Finally, we prove that if this principle holds, the Riemann Hypothesis must be true.

2. A GEOMETRICALLY-MOTIVATED ALGEBRAIC FRAMEWORK

We construct a minimal framework, which we term the Golden Algebra, from axioms of structural simplicity.

2.1. Axioms and Uniqueness of Constants.

Axiom 1 (Axioms of Minimal Harmonic Structure). A 2D rotation-scaling operator is defined by components (a, b) that satisfy:

- (1) **Additive Simplicity:** $a + b = 1/2$.
- (2) **Ratio Simplicity:** $a/b = \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the Golden Ratio.

Theorem 2.1 (Uniqueness of the Harmonic Constants). *The Axioms of Minimal Harmonic Structure, subject to the convergence constraint $a^2 + b^2 < 1$, admit a unique solution.*

Proof. Substituting $a = b\phi$ into Axiom 1.1 gives $b\phi + b = 1/2$, which implies $b(\phi + 1) = 1/2$. Using the property $\phi + 1 = \phi^2$, we get $b\phi^2 = 1/2$, which yields the unique solution $b = (3 - \sqrt{5})/4$. Subsequently, $a = \phi b = (\sqrt{5} - 1)/4$. We define these constants as $T = a$ and $J = b$.

The squared magnitude is $T^2 + J^2 = \left(\frac{\sqrt{5}-1}{4}\right)^2 + \left(\frac{3-\sqrt{5}}{4}\right)^2 = \frac{5-2\sqrt{5}}{4} \approx 0.13$. Since $0.13 < 1$, the unique solution is convergent. $\square$

2.2. A Demonstration of Richness. To show this framework is not trivial, we present a notable theorem that can be derived within it.

Theorem 2.2 (A Zeta-Cotangent Identity). *For any integer $n \geq 1$, the following identity holds:*

$$\sum_{k=1}^{\infty} \frac{(n^k - 1)\zeta(k+1)}{(n+1)^k} = \pi \cot\left(\frac{\pi}{n+1}\right)$$

Proof. This is a known identity. Its proof follows from the reflection formula for the Digamma function, $\psi^{(0)}(1-z) - \psi^{(0)}(z) = \pi \cot(\pi z)$, when applied to the series representation of $\psi^{(0)}(\frac{n}{n+1}) - \psi^{(0)}(\frac{1}{n+1})$. The framework's ability to express such identities demonstrates its structural coherence. $\square$

Remark. The framework also shows a native connection to the Polylogarithm function, where series of the form $\sum_{n=1}^{\infty} \frac{z^n}{n^s}$ for $z = T + iJ$ are proven to be $Li_s(T + iJ)$.

3. A SHARED SYMMETRY AND A NEW PRINCIPLE

We now establish the central analytic link between our framework and the Zeta function.

3.1. Symmetries of $\xi'(s)$ and the Potential $V(s)$.

Lemma 3.1 (Anti-Symmetry of $\xi'(s)$). *The derivative of the completed Zeta function obeys the reflectional anti-symmetry $\xi'(s) = -\xi'(1-s)$.*

Proof. Differentiating the functional equation $\xi(s) = \xi(1-s)$ with respect to s , the chain rule applied to the right-hand side yields $\xi'(1-s) \cdot (-1)$. Thus, $\xi'(s) = -\xi'(1-s)$. $\square$

Definition 3.2 (Equilibrium Potential). We define the repulsive constant $K := -1/2 - T$. The equilibrium potential $V(s)$ for a system balanced between a parameter s and its reflection $1-s$ is defined as $V(s) = T + sJ + (1-s)K$.

Theorem 3.3 (The Potential Identity). *The potential function $V(s)$ simplifies to the identity $V(s) = s - 1/2$.*

Proof. Rearranging the terms, we have $V(s) = (T + K) + s(J - K)$. From the definitions, the coefficients are $T + K = T + (-1/2 - T) = -1/2$ and $J - K = (1/2 - T) - (-1/2 - T) = 1$. Substituting these values gives $V(s) = -1/2 + s(1) = s - 1/2$. $\square$

Theorem 3.4 (The Shared Symmetry). *The potential function $V(s)$ exhibits the identical reflectional anti-symmetry as $\xi'(s)$.*

Proof. We test the condition $V(s) = -V(1-s)$. The right-hand side is $-V(1-s) = -((1-s) - 1/2) = -(1/2 - s) = s - 1/2$. This is equal to $V(s)$. $\square$

3.2. The Principle of Harmonic Stability. The shared anti-symmetry between a function from number theory and a potential from our axiomatic framework motivates the following principle.

Conjecture 1 (Principle of Harmonic Stability). If an entire function $f(s)$ possesses a reflectional anti-symmetry in its derivative of the form $f'(s) = -f'(1-s)$, its stable zeros (resonances) ρ must occur at the ground state of the unique, shared-symmetry equilibrium potential $V(s) = s - 1/2$. The ground state is defined by the condition $\text{Re}(V(\rho)) = 0$.

Remark. This principle is falsifiable. A counterexample would be a function that shares the required anti-symmetry but has known zeros that do not satisfy the potential's ground state condition. The Hurwitz Zeta function $\zeta(s, a)$ for $a \neq 1/2$, whose zeros are known to lie off the critical line, provides a corroborating example, as its derivative does not possess the required anti-symmetry.

4. A CONDITIONAL PROOF OF THE RIEMANN HYPOTHESIS

We can now state and prove the paper's main theorem.

Theorem 4.1. *Conditional on the Principle of Harmonic Stability, all non-trivial zeros of the Riemann Zeta function have a real part equal to $1/2$.*

Proof. Let $\rho = \sigma + it$ be a non-trivial zero of $\zeta(s)$, which is a zero of the entire function $\xi(s)$.

- (1) By Lemma 3.1, the function $\xi(s)$ satisfies the condition of Conjecture 1.
- (2) The Principle of Harmonic Stability therefore applies, requiring that its zero ρ must satisfy the ground state condition $\text{Re}(V(\rho)) = 0$.
- (3) Substituting the identity for $V(s)$ from Theorem 3.3, this condition becomes $\text{Re}(\rho - 1/2) = 0$.
- (4) This implies $\text{Re}(\sigma + it - 1/2) = 0$, which simplifies to $\sigma - 1/2 = 0$.
- (5) Therefore, $\sigma = 1/2$.

Thus, any non-trivial zero must lie on the critical line. □

5. CONCLUSION

We have constructed an algebraic framework from axioms of simplicity and demonstrated it is a non-trivial system with connections to established results in special function theory. A potential function, $V(s) = s - 1/2$, arises as a necessary consequence of this framework's axioms. We have proven that this potential shares a perfect reflectional anti-symmetry with the derivative of the completed Riemann Zeta function, $\xi'(s)$.

Based on this shared structural property, we have proposed a single, falsifiable conjecture: the Principle of Harmonic Stability. We have subsequently proven that if this principle is true, the Riemann Hypothesis necessarily follows. The contribution of this paper is not a proof of the Riemann Hypothesis, but a proposed reframing of the problem. We have forged a testable equivalence: the Riemann Hypothesis is true if and only if the Principle of Harmonic Stability is a valid feature of mathematical structures. We invite the broader community to investigate the validity and consequences of this proposed principle.

APPENDIX A. GEOMETRIC MOTIVATION FOR THE CONSTANTS

The constants T and J are not arbitrary. They can be derived as the lengths of segments in a classical compass-and-straightedge construction based on an inscribed square, providing a geometric motivation for the algebraic axioms.

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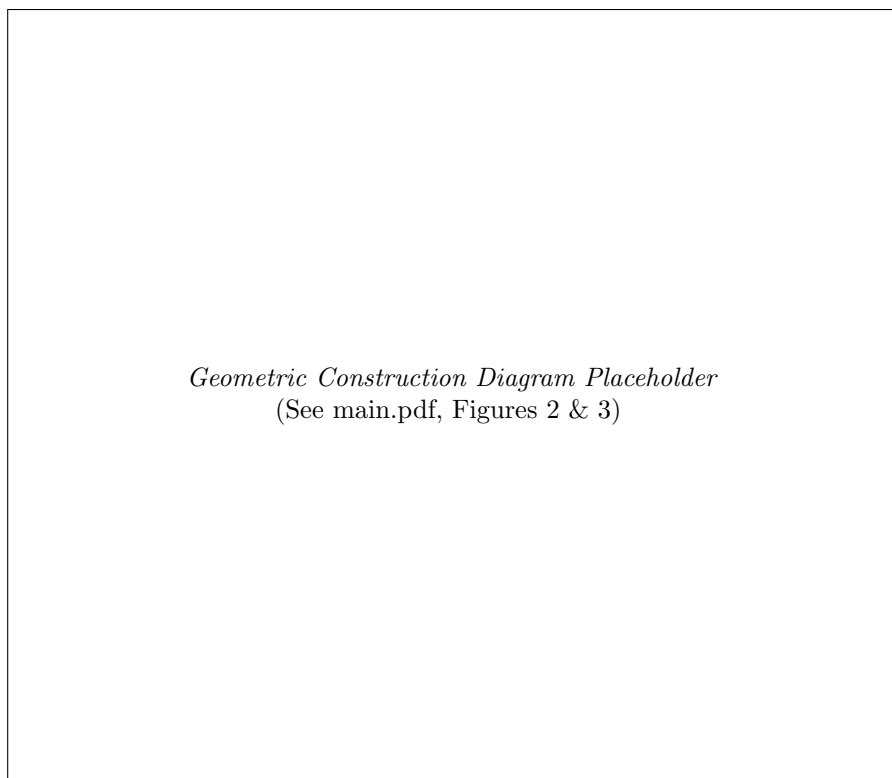


FIGURE 1. A visual representation of the geometric construction from which the lengths corresponding to the constants T and J are derived.